

JAHANVI RANA

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[LinkedIn](#) — [Portfolio](#) — [GitHub](#) — [Kaggle](#)

Professional Summary

Data Analytics and Business Intelligence professional skilled in SQL, Python, Excel, and Power BI, with hands-on experience in data cleaning, EDA, KPI dashboards, customer analytics, and predictive forecasting. Strong foundation in data modeling, visualization, statistical analysis, and data-driven decision making.

Core Competencies

- **Business Intelligence:** Data Analytics, Business Analysis, Product Analytics, Data Visualization
- **Analytics:** Predictive Analytics, Dashboard Development, Data Cleaning, ETL, Data Modeling, Data Pipelines
- **Business Reporting:** KPI Reporting, Forecasting, Statistical Analysis, Decision Support
- **Problem Solving:** Analytical Problem Solving, Performance Analysis, Data Storytelling, Data-Driven Decision Making

Technical Skills

- **Programming:** Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), SQL
- **SQL:** Complex Joins, Aggregations, Subqueries, Views, Filtering, Grouping, Data Transformation
- **Data Analytics:** EDA, Data Cleaning, Data Validation, Data Transformation, Feature Engineering
- **Statistical & Time-Series Analysis:** Statistical Analysis, Time-Series Analysis, Forecasting, Trend Analysis
- **Data Engineering / Pipelines:** ETL, Data Pipelines, Data Validation, Data Preparation, Data Storytelling
- **Business Intelligence:** Power BI, Power Query, DAX, Data Modeling, KPI Dashboards, Business Reporting
- **Data Visualization:** Interactive Dashboard Design, Data Visualization, KPI Development, Drill-throughs
- **Business / Product Analytics:** Customer Segmentation, Retention & Conversion, Revenue Analysis, User Behavior
- **Performance Analytics:** Performance Analysis, Stakeholder Reporting, Data-Driven Decision Making, A/B Testing
- **Advanced Excel:** Pivot Tables, Power Query, VLOOKUP, Conditional Formatting, Data Cleaning
- **Tools & Productivity:** Microsoft Word, Microsoft PowerPoint, Overleaf, Git, GitHub, Google Colab, Kaggle, Jupyter

Projects

1.) Enterprise OTT Streaming Business Intelligence & Customer Analytics Platform k

Tools: Python, Power BI, DAX, Excel, Pandas, Matplotlib

Impact: Built an end-to-end analytics and BI solution to analyze OTT customer engagement, content performance, retention, conversion, and revenue trends.

- Processed and validated raw streaming data using Python (Pandas), performing data cleaning, transformation, and exploratory analytics to analyze customer behavior, engagement, content performance, and revenue trends.
- Designed a multi-page Power BI dashboard covering Executive KPIs, Customer Segmentation, Geography, Content Performance, Revenue, Retention, and Conversion, with dynamic DAX measures for key business metrics.
- Implemented drill-throughs, slicers, bookmarks, and interactive navigation to transform analytical findings into business insights and data-driven decision support.

2.) Gold Market Intelligence & Predictive Forecasting System k

Tools: Python, Excel, Power BI, XGBoost, LightGBM, CatBoost

Impact: Developed a data-driven market intelligence and forecasting system to analyze gold price trends and relationships with macroeconomic indicators across global and Indian markets.

- Collected, cleaned, validated, and transformed multi-source financial datasets using Excel and Python, performing exploratory and time-series analysis and engineering lag, rolling, and other time-series features.
- Developed and compared XGBoost, LightGBM, and CatBoost ensemble models using MAE, RMSE, and R^2 to evaluate forecasting performance and identify market trends.
- Built Power BI dashboards visualizing price trends, macroeconomic indicators, market relationships, heatmaps, and forecasting results to generate interpretable insights and support data-driven decision making.

Research Publication

Research Co-Author

2025

“Enhancing YOLO with Rotated Bounding Boxes and ResNet50 Backbone for Accurate Orientation-Aware Object Detection,”
Proceedings of the ICICC 2026, hosted on SSRN.
Achieved a 13.5% increase in mAP and an 18% improvement in rotated IoU on the DOTA dataset using a custom circular IoU loss function.

[View Publication]

Education

Chandigarh University
Bachelor of Engineering in Computer Science – Artificial Intelligence & Machine Learning

Mohali, Punjab
2022 – 2026
CGPA: 7.75/10

Triple M Public School
Intermediate – Non-Medical with Computer Science

Hoshiarpur, Punjab
2022
Percentage: 88.8%

SGGS Sr. Sec. Public School
High School / Matriculation

Mukerian, Punjab
2020
Percentage: 75%

Training & Certifications

- **Python Essentials 1 & 2** – Cisco Networking Academy & OpenEDG
 - **Preparing Data for Analysis with Microsoft Excel** – Microsoft Learn
 - **Extract, Transform and Load Data in Power BI** – Microsoft Learn
 - **Data Modeling in Power BI** – Microsoft Learn
- Jun 2023

Verify
- Jul 2025

Verify
- Sep 2025

Verify
- Jan 2026

Verify